

Car Price Prediction using Linear Regression & Lasso Regression

Project Report

1. Introduction

The price of used cars depends on multiple factors such as brand, age, fuel type, transmission, and kilometers driven. Predicting car prices accurately is important for both buyers and sellers in the automobile market.

This project focuses on predicting used car selling prices using Machine Learning regression techniques. Linear Regression and Lasso Regression models are implemented to analyze the influence of different car features on selling price.

The project includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and feature importance analysis.

2. Objective

The main objectives of this project are:

- To analyze factors affecting used car prices
 - To perform data preprocessing and feature engineering
 - To build regression models for price prediction
 - To compare Linear Regression and Lasso Regression
 - To identify important features influencing car price
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3. Dataset Description

Dataset Used:

`car_data.csv`

Features:

- **Car Name** → Name and brand of the car
- **Year** → Manufacturing year of the car

- **Selling Price** → Selling price of the car
- **Kilometers Driven** → Total distance driven
- **Fuel Type** → Petrol, Diesel, or CNG
- **Seller Type** → Dealer or Individual
- **Transmission** → Manual or Automatic
- **Owner** → Number of previous owners

Target Variable:

- **Selling Price**

The dataset contains information about used cars and their selling prices.

4. Libraries Used

The following Python libraries were used:

- NumPy
 - Pandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
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5. Methodology

5.1 Import Libraries

Required libraries for data analysis, visualization, and machine learning were imported.

5.2 Load Dataset

The dataset was loaded using Pandas and inspected for:

- Shape
- Missing values
- Data types
- Statistical summary

5.3 Data Cleaning and Feature Engineering

The following preprocessing steps were performed:

- Extracted car brand from car name
- Removed unnecessary columns
- Converted manufacturing year into car age
- Handled categorical variables

5.4 Exploratory Data Analysis (EDA)

Visual analysis was performed using:

- Selling price distribution plots
- Car age vs selling price
- Kilometers driven vs selling price
- Correlation heatmap

EDA helped understand relationships between features and target variable.

5.5 Encoding Categorical Variables

Categorical variables were converted into numerical format using:

- One-Hot Encoding (`pd.get_dummies()`)

5.6 Train-Test Split

The dataset was divided into:

- Training Data → 80%
- Testing Data → 20%

5.7 Feature Scaling

StandardScaler was applied to normalize feature values and improve model performance.

5.8 Model Training

Two regression models were trained:

Linear Regression

A basic regression model that predicts continuous values based on linear relationships.

Lasso Regression

A regression model with L1 regularization that reduces less important feature coefficients.

5.9 Prediction

Predictions were generated using both trained models.

5.10 Model Evaluation

Models were evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

5.11 Residual Analysis

Residual plots were analyzed to check:

- Error distribution
- Model reliability
- Prediction consistency

5.12 Feature Importance

Lasso Regression coefficients were analyzed to identify the most important features affecting car price.

5.13 Overfitting Analysis

Train and test scores were compared to evaluate model generalization and detect overfitting.

6. Regression Models Used

6.1 Linear Regression

Linear Regression establishes a relationship between independent variables and selling price.

Advantages:

- Simple and interpretable
 - Fast training
 - Effective baseline model
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6.2 Lasso Regression

Lasso Regression applies L1 regularization to reduce unnecessary feature influence.

Advantages:

- Performs feature selection
 - Reduces overfitting
 - Improves model simplicity
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7. Evaluation Metrics

Mean Absolute Error (MAE)

Measures average prediction error.

Lower MAE indicates better model accuracy.

Root Mean Squared Error (RMSE)

Measures square root of average squared prediction error.

Lower RMSE indicates better prediction performance.

R² Score

Measures how well the model explains target variance.

Higher R² score indicates better model performance.

8. Results and Analysis

- Linear Regression performed effectively as a baseline model.
 - Lasso Regression reduced less important feature coefficients.
 - Car age negatively affected selling price.
 - Higher kilometers driven reduced resale value.
 - Premium car brands maintained higher prices.
 - Fuel type and transmission significantly influenced car value.
 - Residual analysis indicated stable model performance.
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9. Key Insights

- Older cars generally have lower resale prices.

- Cars with high kilometers driven tend to have reduced value.
 - Automatic transmission cars often have higher prices.
 - Premium brands maintain stronger resale value.
 - Lasso Regression helps reduce overfitting and select important features.
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10. Conclusion

This project successfully developed Machine Learning regression models for predicting used car prices.

Linear Regression provided strong baseline performance, while Lasso Regression improved feature selection and reduced unnecessary feature influence through regularization.

The project demonstrates how Machine Learning can assist in understanding market trends and predicting used car prices accurately.

11. Future Enhancements

Future improvements may include:

- Using larger real-world datasets
 - Applying advanced regression algorithms
 - Hyperparameter tuning
 - Feature engineering
 - Deploying the model as a web application
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12. Project Structure

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OIBSIP_CarPricePrediction/  
├── Car_Price_Prediction.ipynb  
├── car_data.csv  
└── README.md
```

13. References

- Scikit-learn Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation

- Machine Learning Regression Concepts
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Outcome

Successfully built and evaluated Machine Learning models capable of predicting used car prices with good accuracy and meaningful business insights.